

NR7Y CW Firmware - Paddle Rework

Important Note: because of the pins this rework reuses, when the firmware is in CW mode and an input mode that uses the port (PTT+port or any tip/ring mode), the serial port will NOT work correctly. This only affects CHIRP programming while the main firmware is running, it doesn't affect flashing from the bootloader. To use a memory programmer, change out of CW or out of a port input mode.

Other Important Note: external mics (specifically the external mic PTT switch) will no longer work, ever. The internal mic will continue to operate correctly, when the paddle is unplugged. With the paddle plugged in, the mic is switched out of the circuit, even in non-CW modulation modes. You could still plug in and speak into an external mic, but you'd have to use the main PTT button to key up.

These instructions cover the rework variations for UV-K5 v1 (original), UV-K5 v3, and UV-K1.

Un-reworking the original Straight Key mod

If this radio has the older straight key mod, remove the wire. The resistor that was removed for that mod also needs to be removed for this rework, so you're already 1/4 done!

Testing after Rework

The firmware will not allow setting of the input mode to the +port or tip/ring setting before the rework is completed. If possible, while the radio is still open on the benchtop after rework, it's strongly suggested to install a battery and set a tip/ring setting, then test a paddle in the port.

Rework Steps

1. To open the chassis before starting the rework:
 - a. Remove the battery, volume knob (pull straight up), and antenna
 - b. At the back of the radio south of the label, insert a flat blade screwdriver between the metal case and the plastic chassis, then pry down (levering the metal piece up).
 - c. Once the lower end of the metal casing is clear of the plastic chassis, pull it out and down to slide the top end with the volume knob out of the upper section of the chassis. Don't pull too far as the speaker is attached to the plastic front with short wires.
 - d. Pivot the metal casing with the circuit board to the left like opening a book. Take note of the flat plastic shim that goes around the mic/speaker ports, it usually falls off as you open the case.
 - e. With the circuit board facing up on the workbench, observe the display is attached to the board using a black plastic retaining frame. At the lower left

corner leg of the display frame (closest to the Side 1 / Side 2 buttons), shim a small flat bladed screwdriver or plastic pry tool into the rectangular hole in the circuit board to the outer side of the leg. Pry back, pushing the leg clip toward the middle of the radio to release it, then lift the clip out of its rectangular hole. With that corner out, the display will be loose enough to jiggle and lift up slightly, then carefully flip it up over the top of the circuit board.

2. Remove these two resistors: the “straight key mod” resistor on the left, and the resistor on the right between the two plastic port alignment pins. Note their position near the right edge of the board and below the mic/speaker ports.

Image: Remove these two resistors from the V1 board

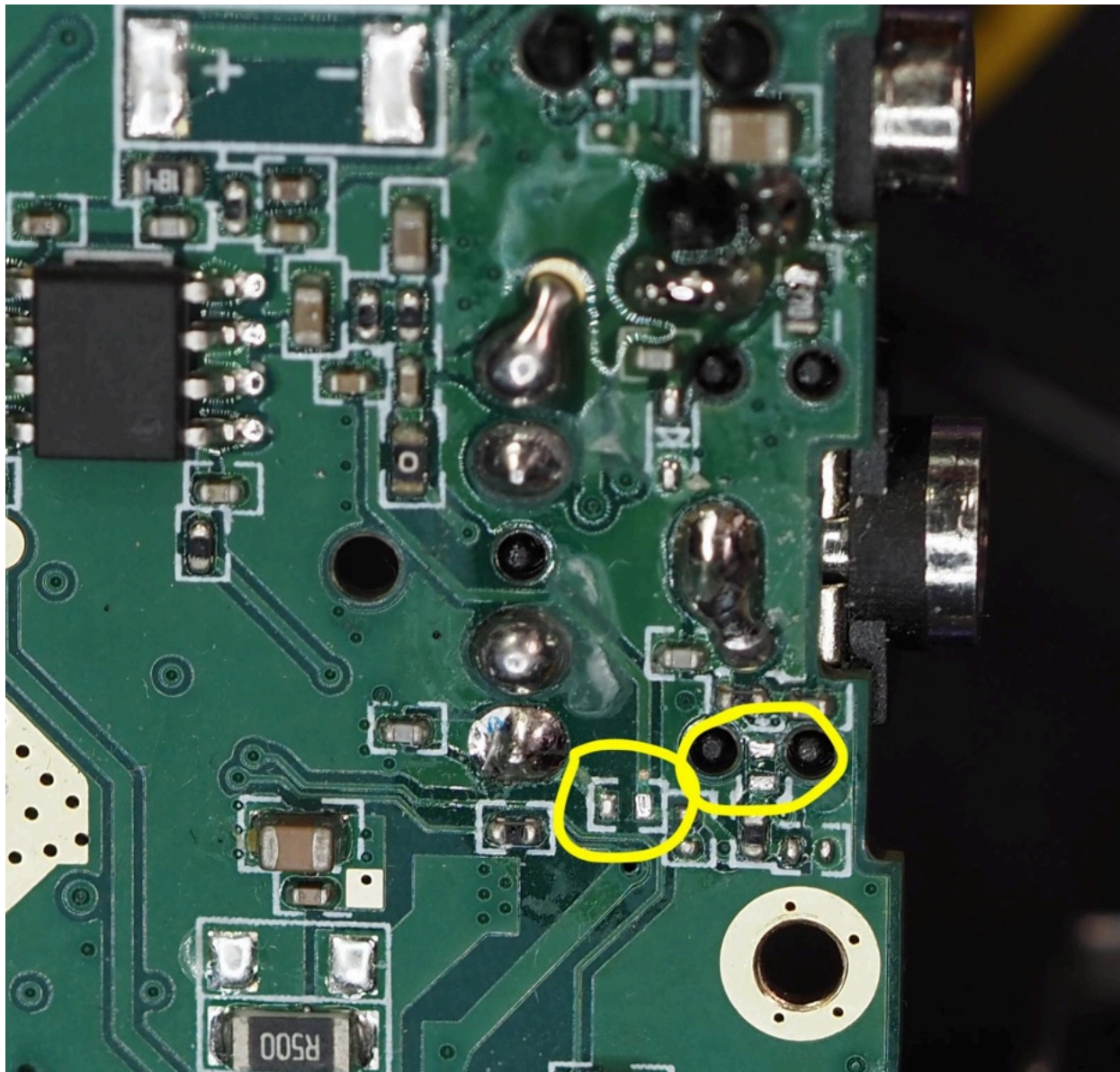


Image: The resistors to remove are in the same position in a UVK5v3

Note: The circuit board is labeled “V2.0” inside a v3 radio

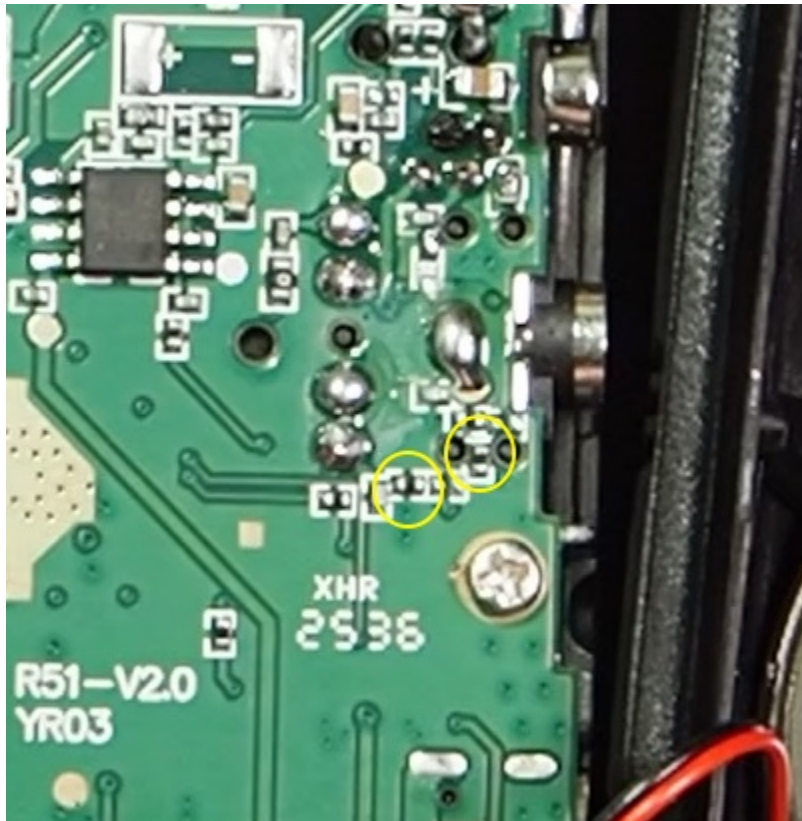
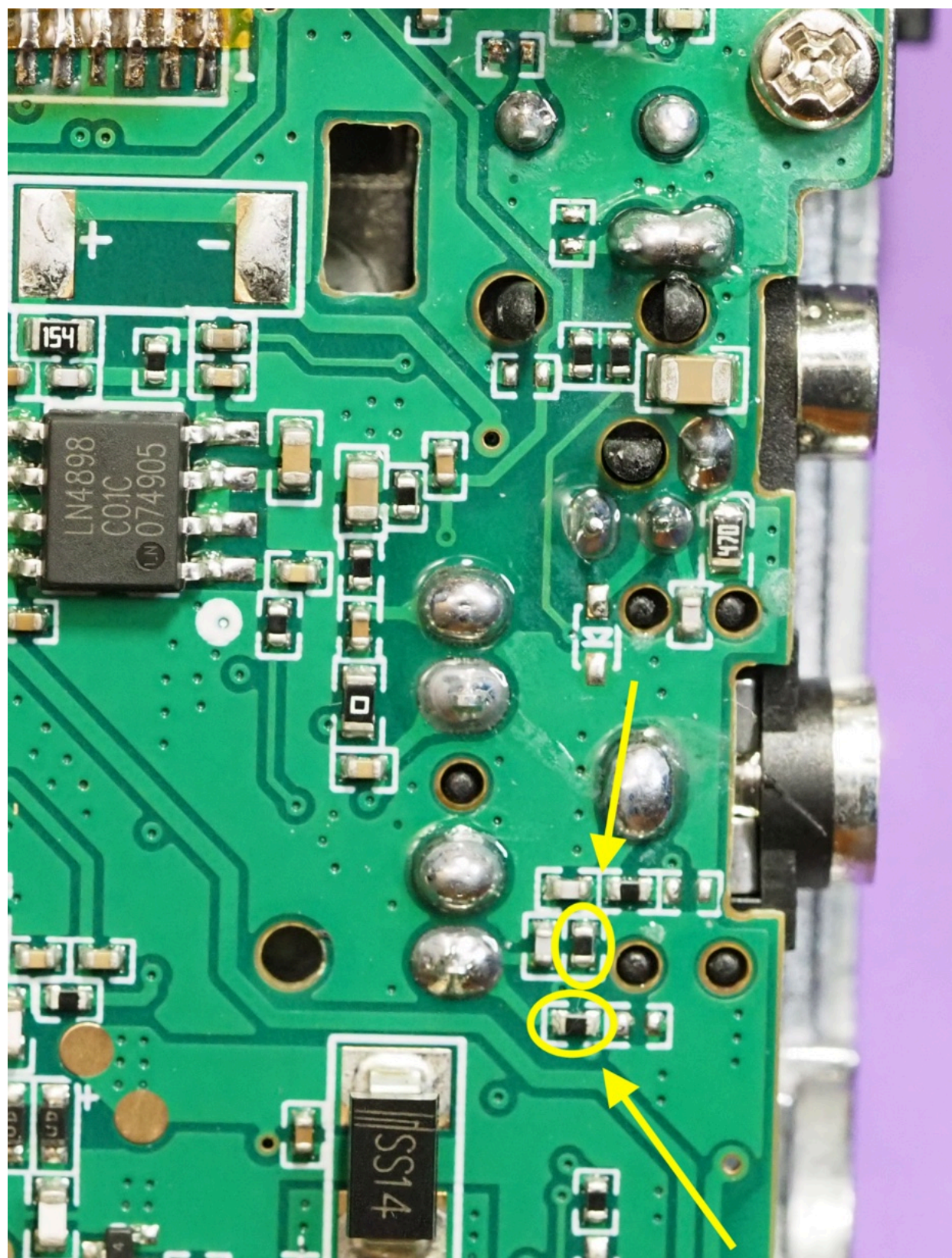


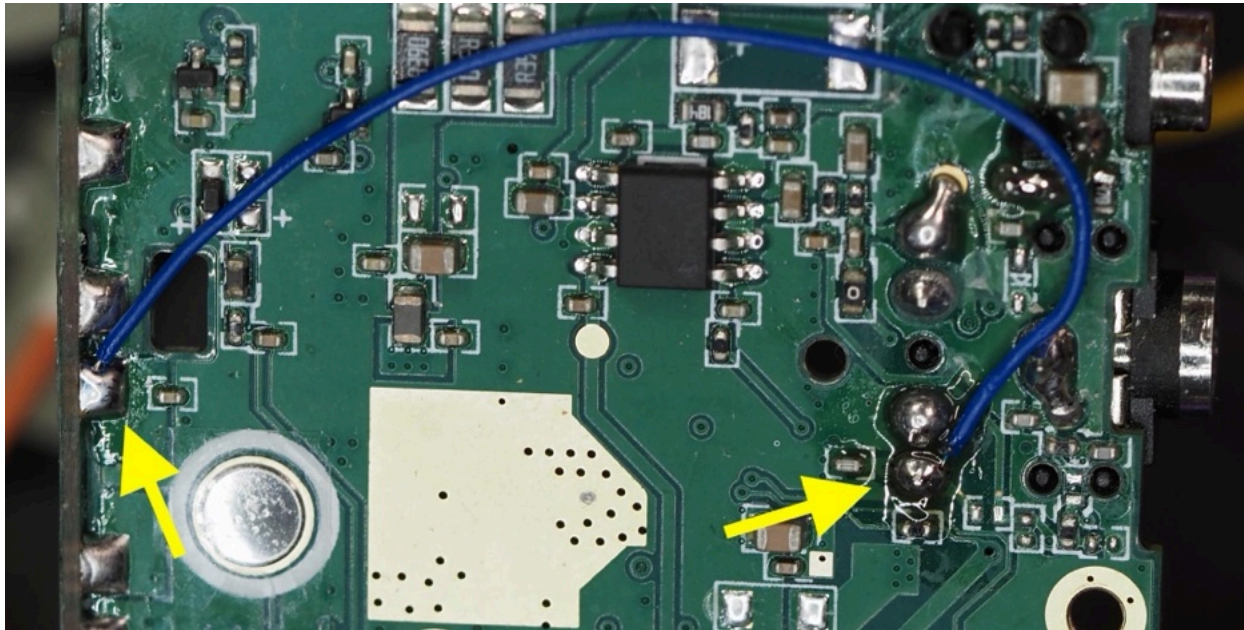
Image: inside the K1, the resistors have moved slightly



3. Add a wire between the PTT pin on the left side, to the 3.5mm Tip terminal on the right. Size of the wire and flushness to the board does not matter, as this wire sits under the display.

On the left end of the wire, keep it out of the way of the screen retaining bracket that goes through the rectangular hole - in this picture the hole is partially covered by the blue wire, and it will need to be pulled up and left before reinstalling the screen.

Image: connect the indicated terminals with a wire



4. **Note - this version of the rework only works properly on firmware beta6 and above.** Add a wire between the 3.5mm Sleeve terminal, down to the third test pad below the microphone. It's best to route the wire to the right of the mic and speaker terminals, then up to the Sleeve terminal. This helps avoid the rework wire getting under any of the key pad buttons. Aside from this, the path the wire takes isn't important.

"Hey isn't that the SWDIO line for SWD programming and debug?" - yes it is, and this rework will prevent SWD from working (even in the bootloader). If you need to use SWD debugging, you'll need to remove the wire from the SWDIO pad or perform the older beta rework instead (see the bottom of this doc). If you have no idea what SWD is, don't worry about this.

Image: Connect the indicated terminals with a wire

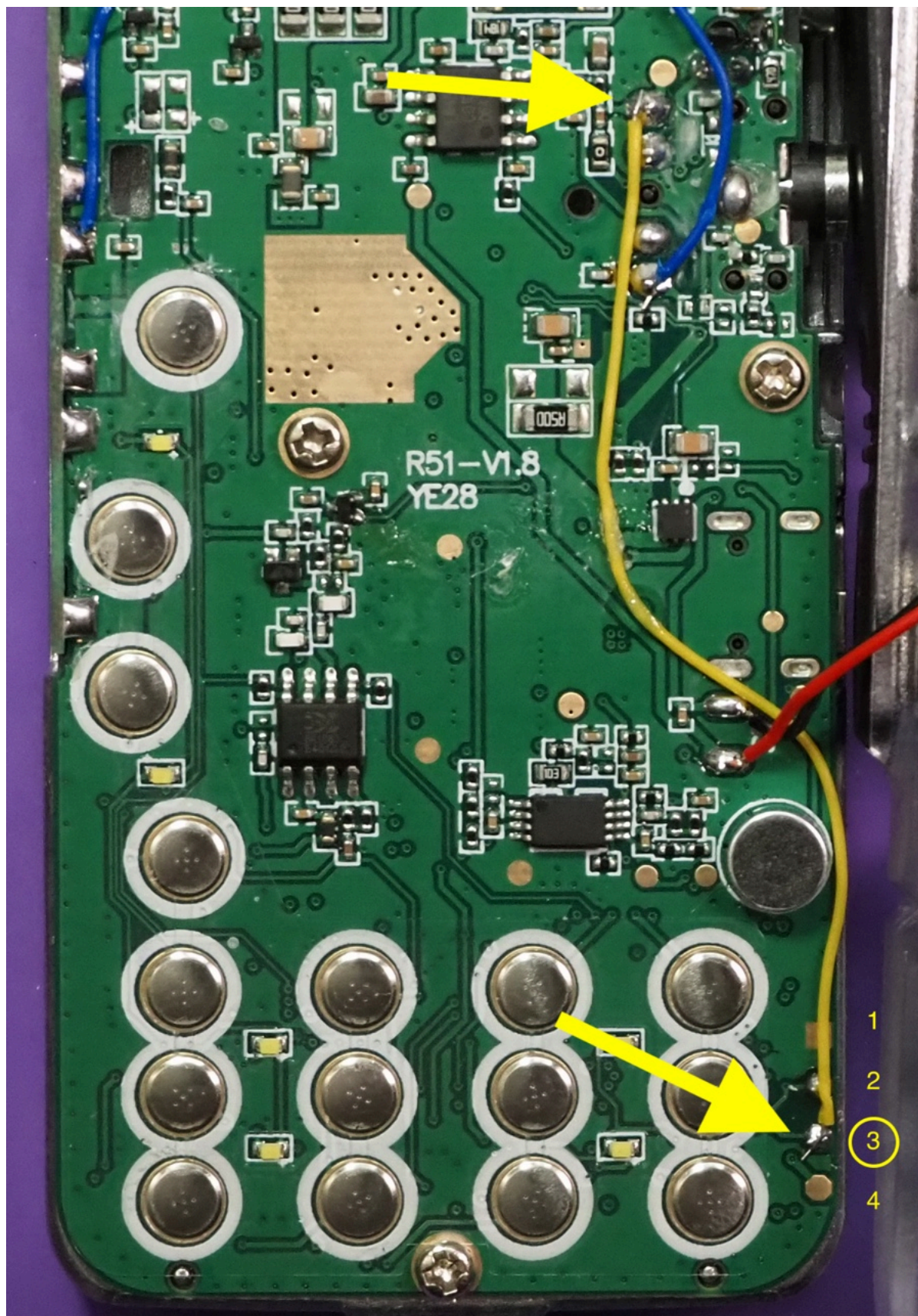


Image: The completed rework on a v3 radio looks almost identical.

This rework was performed using wires stripped out of the USB cable that comes with the radio.

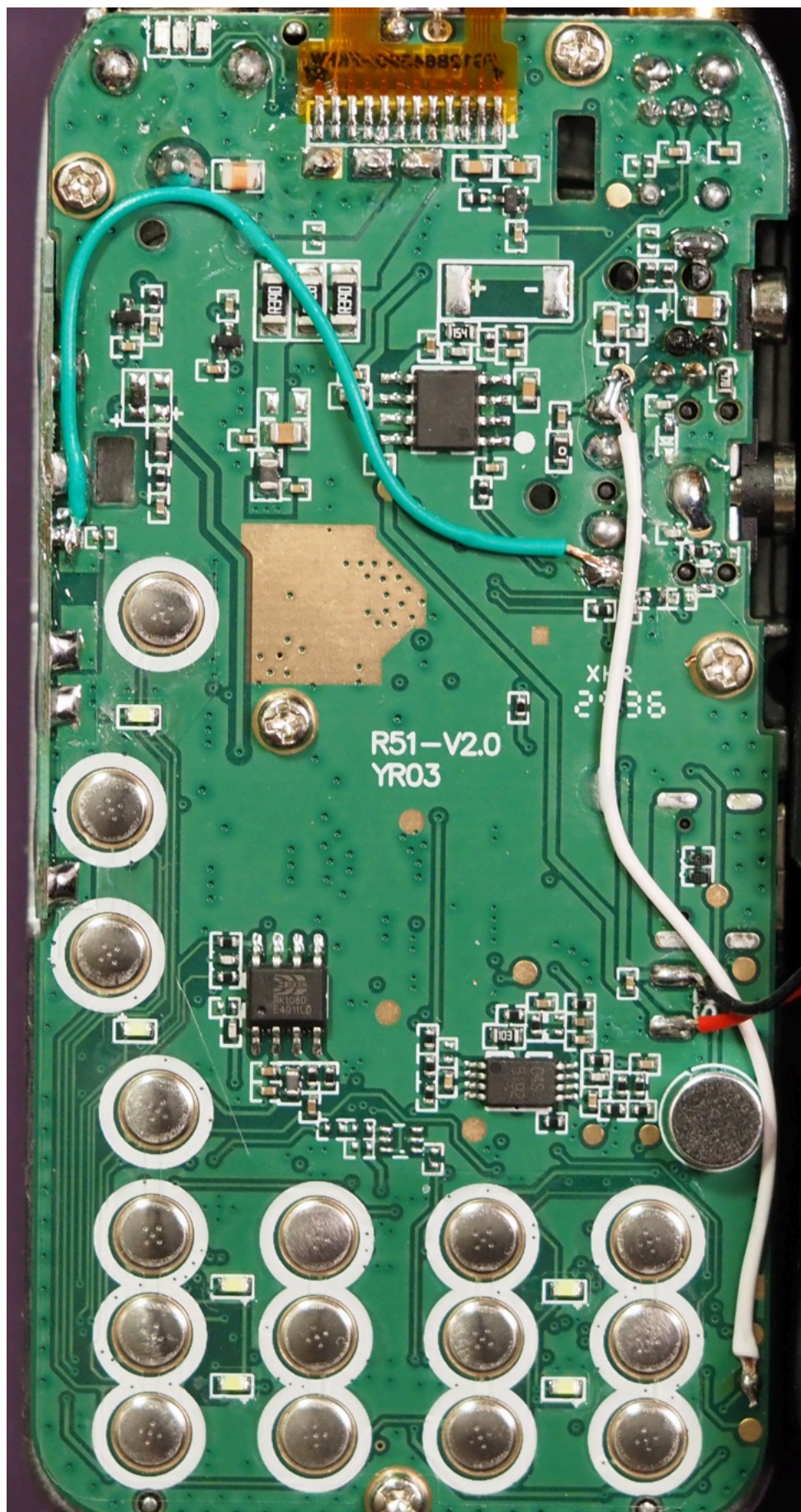
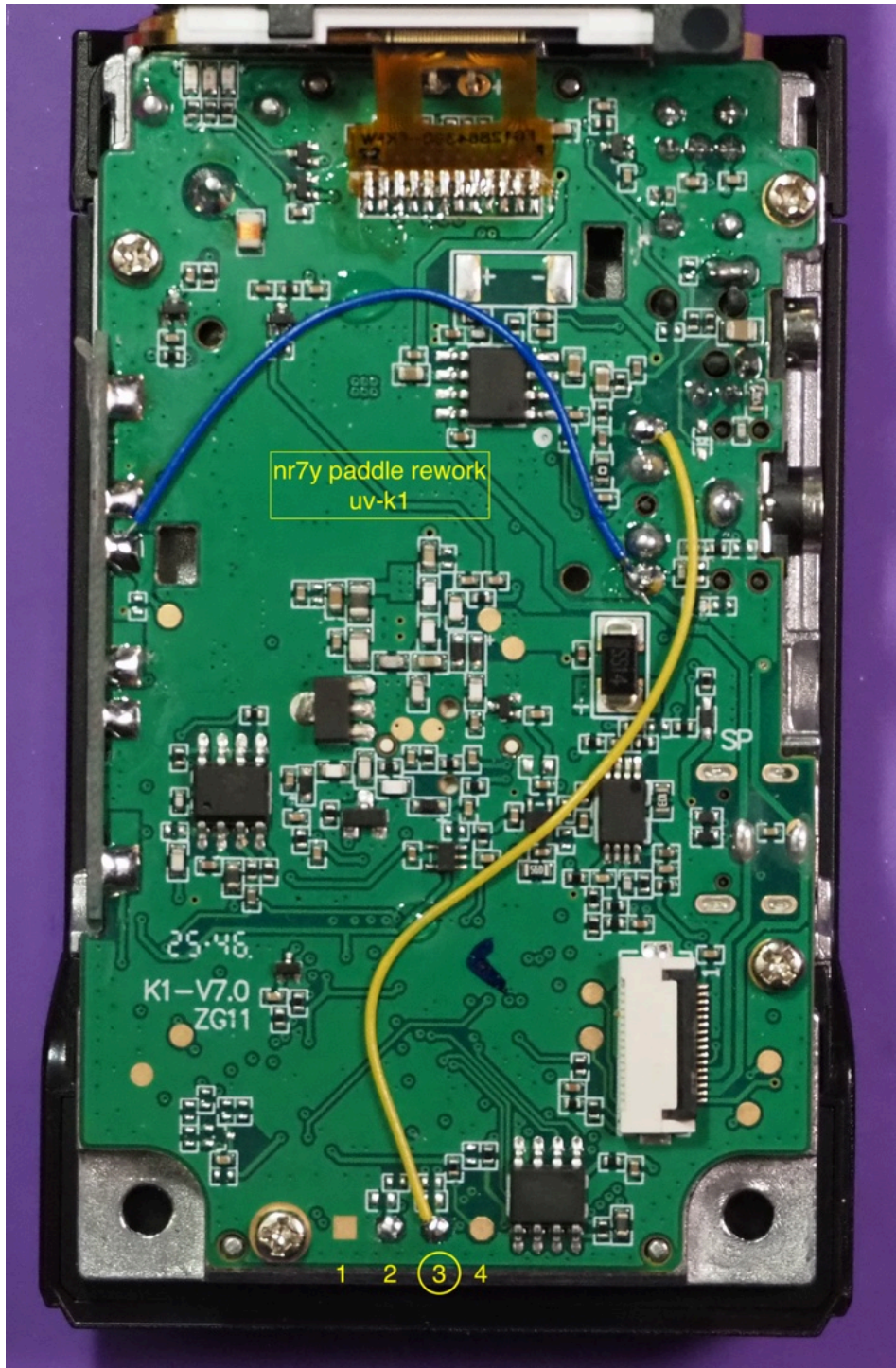


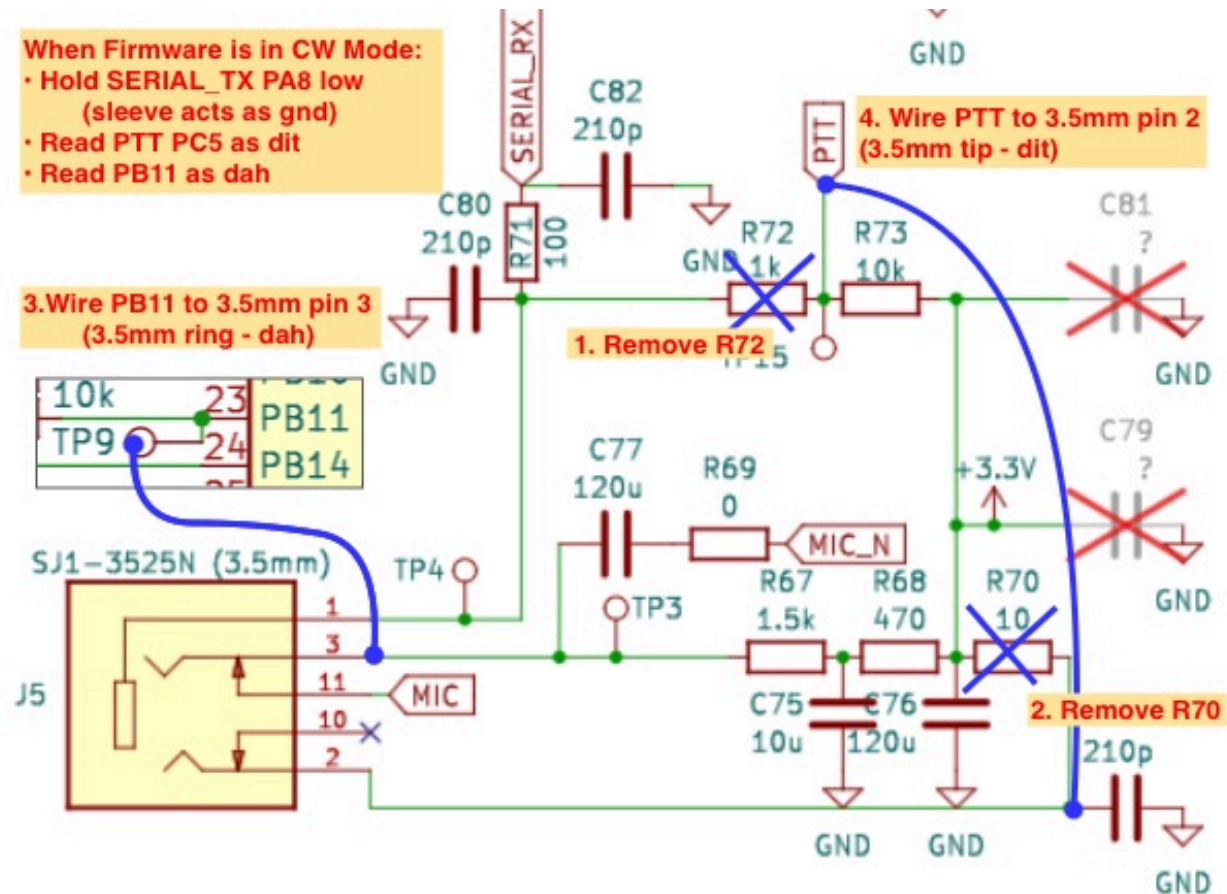
Image: rework wire on a K1 is also the third terminal starting from the square



5. With the rework complete, test the paddle input, then reassemble the radio. Take note of the display condition and inside of the clear plastic display cover, as they may need polishing to eliminate fingerprints or debris. Don't forget to reinstall the flat plastic piece

around the ports before sliding the inner casing back into the plastic chassis. Press down hard on the label to snap the casing back into place.

The rework from the schematic level:



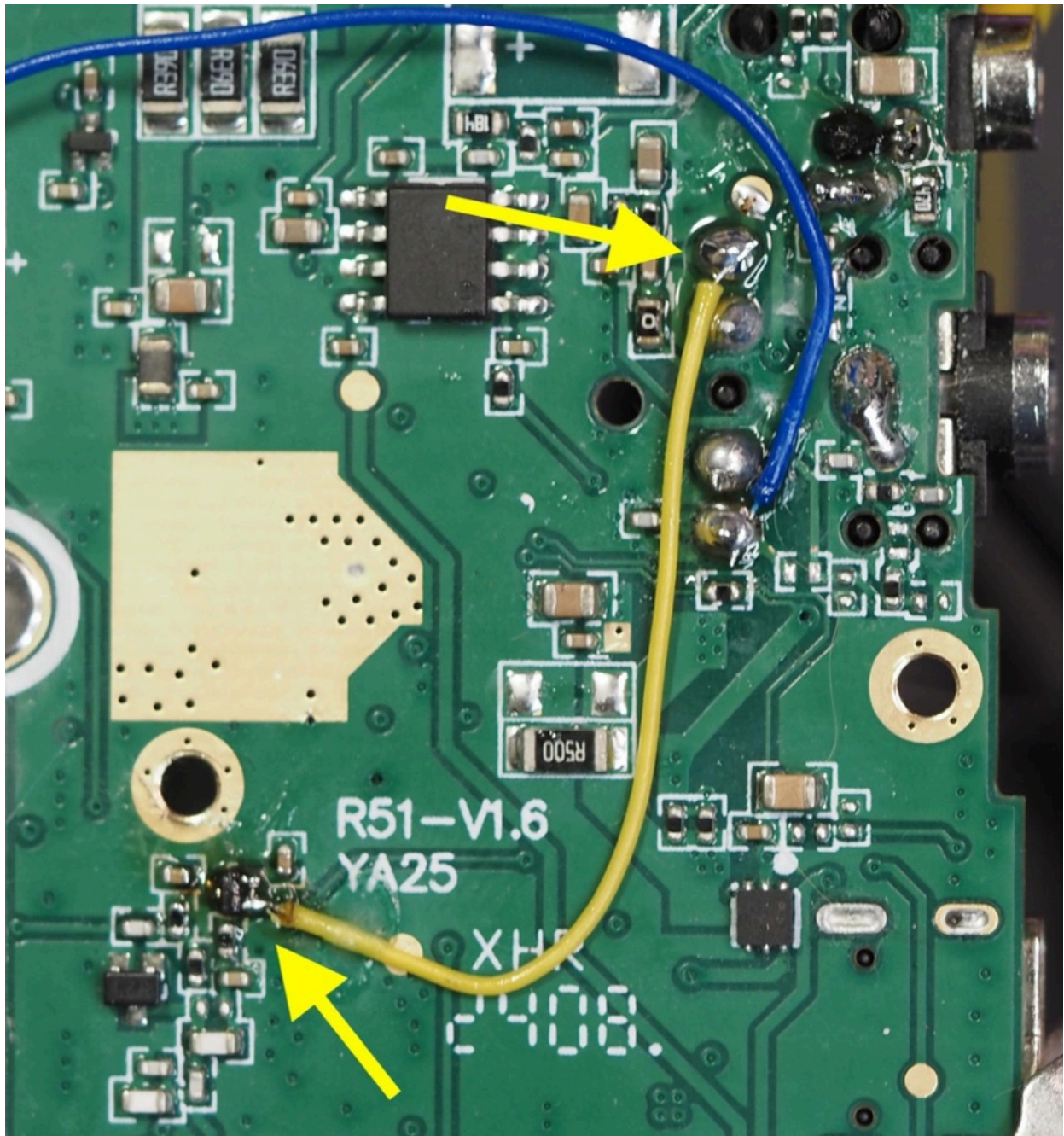
Original schematic by MentalDetector:

https://github.com/mentalDetector/Quansheng_UV-K5_PCB_R51-V1.4_PCB_Reversing_Rev_0.9?tab=readme-ov-file

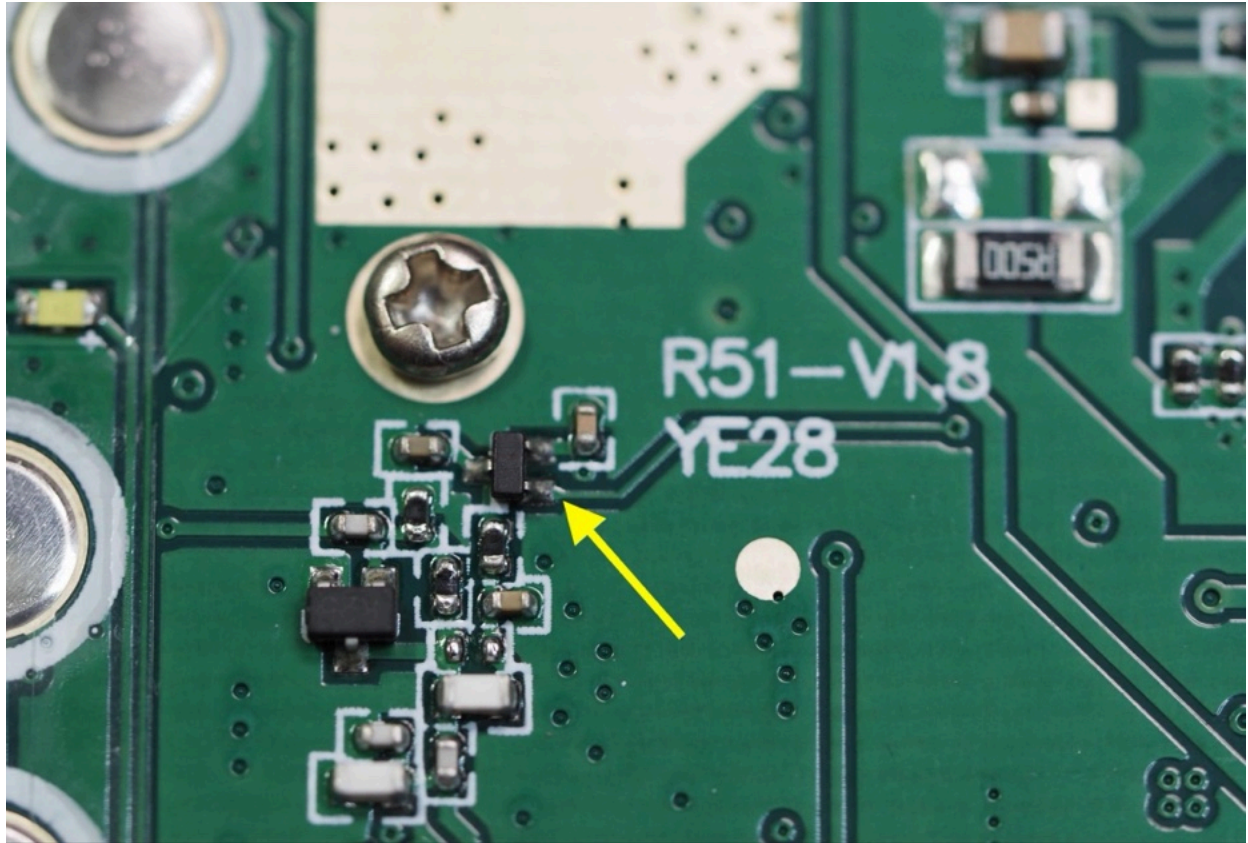
----- older BETA instructions -----

NOTE: if your radio has this rework, and you don't want to change to follow the new instructions, use the "ring b15" firmware binary. Otherwise, the ring input will not work correctly, because firmware will try to read it from the new pin instead.

6. BETA INSTRUCTION - THIS IS CHANGING - Add a wire between the 3.5mm Sleeve terminal, and the tiny transistor leg. Sorry about the tiny connecting point. Using solder flux and a magnifier will help.



Closeup of the transistor leg where the wire needs to be connected:

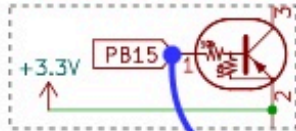


OLD BETA REWORK schematic:

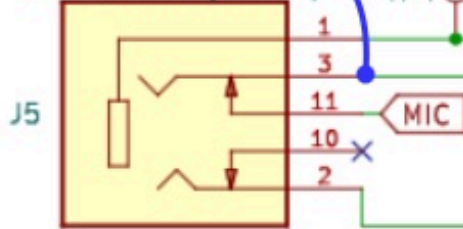
When Firmware is in CW Mode:

- Hold SERIAL_TX PA8 low (sleeve acts as gnd)
- Read PTT PC5 as dit
- Read PB15 as dah

3. Wire PB15 to 3.5mm pin 3 (3.5mm ring - dah)



SJ1-3525N (3.5mm)



1. Remove R72

4. Wire PTT to 3.5mm pin 2 (3.5mm tip - dit)

2. Remove R70

